

PHY302: 1st Mid-Semester Examination (20 points)

Instructor: Manabendra Nath Bera
(Dated: 2 September 2025, 3:00 PM - 4:00 PM)

1. (i) For a Hermitian operator $M = M^\dagger$ acting on a complex Hilbert space $\mathcal{H}$, prove that

$$\langle \phi | M | \phi \rangle \in \mathbb{R},$$

for any vector $|\phi\rangle \in \mathcal{H}$. (5 points)

- (ii) Consider a Hermitian operator $N = \sum_{i=1}^d n_i |e_i\rangle\langle e_i|$. Prove that

$$\langle e_i | e_j \rangle = 0,$$

for $n_i \neq n_j$. (3 points)

- (iii) Prove set of all $(d \times d)$ -Hermitian operators forms a real vector space. (2 points)

2. Consider Hermitian matrices M_1, M_2, M_3, M_4 that obey

$$M_i M_j + M_j M_i = 2 \mathbb{I} \delta_{ij}.$$

- (i) Show that the eigenvalues of M_i are ± 1 . (2 points)

- (ii) Show that M_i are traceless. (2 points)

- (iii) Show that M_i can not be odd-dimensional matrices. (1 points)

3. Consider a quantum particle moving in a one-dimensional potential that is an even function of x , meaning $V(x) = V(-x)$.

Show that the energy eigenstates of the particle can be chosen to be either even or odd under $x \rightarrow -x$. (5 points)

PHY302: 2nd Mid-Semester Examination (30 points)

Instructor: Manabendra Nath Bera

1. (a) Derive an uncertainty relation for the spin-1/2 operators $\hat{S}_x$ and $\hat{S}_y$. That is, given a state $|\psi\rangle$, derive the relation of the form

$$(\Delta \hat{S}_x)(\Delta \hat{S}_y) \geq f(\hat{S}_x, \hat{S}_y, |\psi\rangle),$$

and express the function $f(\hat{S}_x, \hat{S}_y, |\psi\rangle)$. (4 points)

- (b) Consider a general spin state $|n; +\rangle$ described in terms of the angles θ and ϕ . Calculate $\Delta \hat{S}_x$ and $\Delta \hat{S}_y$. (4 points)

2. A particle is moving in a box divided into left and right compartments by a thin partition. The normalized vector $|l\rangle$ ($|r\rangle$) represents the state of the particle if it is on the left (right) compartment and satisfies $\langle l|r\rangle = 0$. Consider a state of the particle at time $t = 0$,

$$|\psi(0)\rangle = \alpha|r\rangle + \beta|l\rangle,$$

where $|\alpha|^2 + |\beta|^2 = 1$. The particle can tunnel through the partition and is characterized by the Hamiltonian

$$\hat{H} = \delta(|r\rangle\langle l| + |l\rangle\langle r|),$$

where δ is a real number with the dimension of energy.

- (a) Find the state of the particle $|\psi(t)\rangle$ at time $t = t$. (2 points)

- (b) Suppose at $t = 0$ the particle is in the right compartment with certainty. What is the probability of finding the particle in the left compartment as a function of time? (2 points)

- (c) Suppose a printer made an error and wrote H as

$$\hat{H} = \delta |l\rangle\langle r|.$$

Prove that the probability conservation is violated. (3 points)

3. (a) For a one-dimensional quantum simple harmonic oscillator with mass m and angular frequency ω , express the creation and annihilation operators in the Heisenberg picture, i.e., $\hat{a}_H^\dagger(t)$ and $\hat{a}_H(t)$, where the annihilation operator in the Schrodinger picture is given by

$$a = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + i \frac{\hat{p}}{m\omega} \right). \quad (5 \text{ points})$$

- (b) Consider a function, known as the correlation function for some state $|\psi\rangle$, defined by

$$C(t) = \langle \psi | \hat{x}_H(t) \hat{x}_H(0) | \psi \rangle,$$

where $\hat{x}_H(t)$ is the position operator in the Heisenberg picture. Evaluate the correlation function explicitly for the first excited state, i.e., $|\psi\rangle = |\phi_1\rangle$, of a one-dimensional quantum simple harmonic oscillator. (10 points)

PHY302: EndSem Examination [50 points]

Instructor: Manabendra Nath Bera

Negative markings:

- (a) Missing name and/or registration number on the answer sheet. [-3 points]
 (b) Writing that is difficult to read or follow. [-3 points]

1. Indicate **T** or **F** for true or false, respectively, with clear justification. [10 points]

(a) Consider any operator A and two arbitrary complete sets of orthonormal bases $\{|\psi_i\rangle\}$ and $\{|\phi_m\rangle\}$, where $\langle\psi_i|\phi_m\rangle \neq 0$. Then, $\sum_i \langle\psi_i|A|\psi_i\rangle \neq \sum_m \langle\phi_m|A|\phi_m\rangle$.

(b) The uncertainty ΔS of a time-independent operator S is time-independent.

(c) For any finite integer $N \geq 2$, there exist $(N \times N)$ -matrices, X and P , such that $[X, P] = iI$.

(d) Let $T \otimes S$ be a linear operator on the vector space $\mathcal{V} \otimes \mathcal{W}$. Then $(T \otimes S)^\dagger = S^\dagger \otimes T^\dagger$.

(e) $\hat{\mathbf{r}} \cdot \hat{\mathbf{p}} = \hat{\mathbf{p}} \cdot \hat{\mathbf{r}} - 3i\hbar I$. Here $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$ are the position and momentum operators in 3-dimension, respectively.

(f) For a simple harmonic oscillator the number operator in the Heisenberg picture is identical to the number operator in Schrödinger picture.

(g) $[\mathbf{L}, \hat{x}^2] = 0$. Here $\mathbf{L}$ is orbital angular momentum operator, and $\hat{x}$ is the position- x operator.

(h) If $\mathbf{J}_1$ and $\mathbf{J}_2$ are sets of operators each satisfying the algebra of angular momentum, the combination $\mathbf{J}_1 - \mathbf{J}_2$ also satisfies the algebra of angular momentum.

(i) Let $\mathbf{J}_1$ and $\mathbf{J}_2$ be angular momentum operators. The operator $\mathbf{J}_1 \cdot \mathbf{J}_2$ commutes with $J_{1x} + J_{2x}$.

(j) The wave function of a particle in a three-dimensional central potential must vanish at the origin.

2. (a) Consider the one-dimensional time-independent Schrödinger equation for some arbitrary potential $V(x)$. Prove that if a solution $\psi(x)$ has the property that $\psi(x) \rightarrow 0$ as $|x| \rightarrow \infty$, then the solution must be non-degenerate and real, apart from a possible overall phase factor. [3+3 points]

(b) A particle of mass m moves in one dimension under the influence of a potential $V(x)$. Suppose it is in an energy eigenstate $\psi(x) = (\gamma^2/\pi)^{1/4} \exp(-\gamma^2 x^2/2)$ with energy $E = \hbar^2 \gamma^2/2m$.

(i) Find the mean position of the particle. [1 point]

(ii) Find the mean momentum of the particle. [1 point]

(iii) Find $V(x)$. [2 points]

3. Consider a one-dimensional simple harmonic oscillator. Do the following algebraically, i.e., without using the explicit forms of the wave-functions.

(a) Construct a state as a linear combination of $|0\rangle$ and $|1\rangle$ such that $\langle \hat{x} \rangle$ is as large as possible. Here $|0\rangle$ and $|1\rangle$ are the ground and first excited states. [3 points]

(b) Suppose the oscillator is in the state constructed in (i) at time $t = 0$. What is the state vector for $t > 0$? [1 point]

(c) Evaluate the expectation value $\langle \hat{x} \rangle$ as a function of time for $t > 0$ using (i) the Schrödinger picture and (ii) the Heisenberg picture independently. [3 points]

(d) Evaluate $\langle (\Delta x)^2 \rangle$ as a function of time using both Schrödinger and Heisenberg pictures. [3 points]

Note, $\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(a + a^\dagger)$, where $a^\dagger$ is the creation operator.

4. (a) A state $|l, m\rangle$ is a simultaneous eigenstate of the orbital angular momentum operators L^2 and L_z :

$$L^2|l, m\rangle = \hbar^2 l(l+1)|l, m\rangle, \quad L_z|l, m\rangle = \hbar m|l, m\rangle.$$

(i) Show that the square of the eigenvalue of L_z cannot exceed the eigenvalue of L^2 . [2 points]

(ii) Calculate $\langle L_x \rangle$ and $\langle L_x^2 \rangle$ for the state $|l, m\rangle$. [3 points]

Hint: You may use the relations $L_\pm = L_x \pm iL_y$ and $L_\pm|l, m\rangle = \hbar \sqrt{l(l+1) - m(m \pm 1)}|l, m \pm 1\rangle$.

(b) Consider a quantum particle in a 3D-potential $V(\mathbf{r})$.

(i) Prove that the rate of change of the expectation value of the orbital angular momentum $\mathbf{L}$ is equal to the expectation value of the torque:

$$\frac{d}{dt}\langle \mathbf{L} \rangle = \langle \mathbf{N} \rangle,$$

where $\mathbf{N} = \mathbf{r} \times (-\nabla V)$. You may use the formula $i\hbar \frac{\partial V}{\partial x_i} = [V, p_i]$. [4 points]

(ii) Calculate $\frac{d}{dt}\langle \mathbf{L} \rangle$ for any spherically symmetric potential and explain your result. [1 point]

5. (a) Consider two non-interacting systems with angular momentum operators $\mathbf{J}_1$ and $\mathbf{J}_2$ with angular momenta $j_1 = \frac{1}{2}$ and $j_2 = \frac{1}{2}$ respectively. The combined system has the total angular momentum j and its z -component m . Here we used the total angular momentum operator as the sum of individual ones, i.e., $\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2$. One may express the angular momentum state (in coupled bases) $|j_1, j_2; j, m\rangle$ of the composite in terms of uncoupled bases $|j_1, j_2; m_1, m_2\rangle$. Here, m_1 and m_2 are the z -component of individual angular momenta $\mathbf{J}_1$ and $\mathbf{J}_2$, respectively, and m is z -component of total angular momentum $\mathbf{J}$.

Find the matrix representation of the angular momentum operators J_x , J_y , and J_z in the coupled bases $|j_1, j_2; j, m\rangle$. (2+2+1 points)

(b) A particle of mass m is constrained to move between two concentric impermeable spheres of radii $r = a$ and $r = b$, with $b > a$. There is no other potential. Find the ground state energy and normalized wave function. [5 points]